

CS 2308: Foundations of Computer Science II

Summer 2025

The course is an introduction to Abstract Data Types (ADTs) including lists, stacks, and queues. Searching and sorting, pointers and dynamic memory allocation, and simple classes and objects also will be covered. The course is a continuation of CS 1428. Prerequisite: CS 1428 with a grade of "C" or better.

FOUNDATIONS OF CS II

San Marcos Campus

S M T W T F S

10:00 AM - 11:40 AM Type: Class Building: Derrick Hall Room: 00234 Start Date: 05/27/2025 End Date: 06/27/2025

Instructor: Keshav Bhandari
207H CML
k_b459@txstate.edu

TA: Dipesh Pandit
rab347@txstate.edu

Office Hrs: Mon/Tue/Wed/Thur/Fri
(Email/Talk First)
(09:00 AM - 10:00 AM)

Available VIA Zoom, Contact him
<https://txstate.zoom.us/my/dipesh12>

Zoom:
<https://txstate.zoom.us/my/meet.keshav>

Contact TA if you need help, setup a meeting

** BRINGING YOUR LAPTOP IN EVERY CLASS CAN BE HELPFUL*

Learning Management System:

- Canvas** - Class materials are available on Canvas (**Use google chrome**)
- Zybooks** - Assignments and Labs (**Use google chrome**)
- TopHat** - Participation and Response system (**Use google chrome**)

FAQs:

1. How to setup zybooks?

1. Access one of the homeworks from Homework modules from canvas
2. Get ZyBooks access code from BookStore if you are enrolled in the BookSmart program, use the access code when prompted. If not, pay to subscribe. Issues: contact support@zybooks.com, Use TXSTATE Email

2. How to set up TopHat?

1. Go to <https://tophat.com/>
2. Create an Account or sign up or use your existing account
3. Join course:

Tutorial: <https://www.youtube.com/watch?v=9sg5krqNAJk>

(Note please access tophat from course-site, from the left navigation bar on Canvas you will find Tophat at the bottom)

Tools: CLION <https://www.jetbrains.com/clion/>

Please install CLION before attending the class.

Use your student email to get a free license.

TextBook: Tony Gaddis, Starting out with C++: From Control Structures through Objects, 9th Edition, ISBN: 0134498372 (8th edition is allowed!). (**Book is not compulsory**)

Classroom Behaviour: Do not disrupt the class and eating inside the classroom is a big NO unless you share your food 😊

Grading:	TopHat Participation	5% + 5%	PopQuiz/Attendance
	Knowledge Tests	5%	
	Homeworks (Zybooks)	20%	Coding
	Projects (1) (Github/Report)	25%	Coding/Reporting/TeamWork
	Quizzes (Canvas-TakeHome)	10% x 2	MCQs Code Analysis
	Final Exam (Paper, FINAL)	20%	MCQs Code Analysis

Tentative Schedules:

Lecture#	L01	L02	L03	L04	L05	L06	L07	L08	L09	L10	L11	L12	L13	L14	L15	L16	L17	L18	L19	L20	L21	L22	L23	L24	L25	L26	L27	L28	L29
Units	Unit 01, Unit 02					Unit 03					Unit 04				Unit 05		Unit 06				Unit 07		Unit 08		Buffer				
Contents	Functions, Organization, Arrays, Vectors, Lists and some comparison Pointers & Dynamic Memory Allocation					Algorithm & Analysis Sorting, Searching, Divide & Conquer, Analysis					Struct, Operator Overloading & Classes				UNIX Prog BASH		Abstract Data Structure				Template		Classes An OO Design		Revision Discussion Fun Coding				

- **ATTENDANCE** You will honestly grade yourself for this as a part of self-evaluation assignment.
- **PARTICIPATION** Bring a web-enabled device to each class to access TopHat. You will be asked questions in some of the classes (2 points each: 1 for correctness, 1 for any answer). 80% of the total possible points earns full credit for participation. This means you can miss up to 20% of the points and still get 100%.
- **QUIZZES** are take-home.
- **Knowledge Tests** consist of 8-10 quiz questions per chapter. These quizzes have unlimited attempts, no time limits, and must be completed by 4/25/2025.
- **FINAL** is in person (**Hard Copy notes allowed, limiting to 4 letter size pages counting both sides**)

- **SCHEDULES:**

<https://www.registrar.txst.edu/persistent-links/final-exam-schedule/final-exam-schedule.html>

POT 51 - Summer I (1st 5 weeks)

10:00 AM - 11:59 AM

Fri. June 27, 2025

11:00 AM-1:30 PM

- **All Projects and Homeworks have same deadline 06/25/2025**
- **Please submit the final version of your project only, only one submission is allowed. Once your project is graded you cannot fix your project for regrading.**

Prerequisites: C or higher in CS 1428: Foundations of Computer Science I

Course Objectives:

At the end of the course, the students should be able to:

1. Describe and demonstrate at least two different algorithms for searching and at least two different algorithms for sorting.
2. Implement a divide-and-conquer algorithm to solve an appropriate problem (binary search).
3. State the time/space efficiency of various algorithms (using one of 6 categories of mathematical functions).
4. List the 6 categories of mathematical functions used in analyzing algorithms in order from slowest to fastest growing.
5. Read and write C++ code that uses pointer variables and memory operations (new, &, *, delete), including pointers to arrays, structures, and objects and the -> operator.
6. Write C++ code that resizes an array using dynamic memory allocation.
7. Write C++ code that deletes dynamically allocated memory to avoid memory leaks.
8. Describe the basic concepts of object-oriented programming.
9. Design, implement, test, and debug simple programs (using objects) in an object-oriented programming language (C++).
10. Describe how the class mechanism supports encapsulation and information hiding.
11. Develop (implement) programs using multiple classes and arrays of objects
12. Develop and use appropriate algorithms, especially for processing lists (insert, remove, search, sort, etc.)
13. Describe structured programming in terms of modules and functions.
14. Develop (implement) programs with source code separated into multiple files, including header (.h) files
15. Create, compile, and run a C++ program in a unix style command-line environment
16. Develop (Implement) C++ programs that create and use simple linked-lists, including code to insert into, delete from, and traverse a linked list structure.
17. Compare and contrast the costs and benefits of dynamic and static data structure implementations.
18. Describe the principle of the Abstract Data Type (ADT) and, in particular, explain the benefits of separation of interface and implementation.
19. Implement user-defined data structures in a high-level language.
20. Implement the list, stack, and queue ADT using arrays and linked lists.
21. Write programs that use each of these data structures: linked lists, stacks, and queues.

Makeup Policy:

Missed participation (TopH and daily quizzes) and TESTs cannot be made up.

Notifications from the instructor:

Notifications related to this class will be either:

- sent to your Texas State e-mail account (be sure to check it regularly) OR
- posted in Canvas under Announcements (set up your Canvas notifications properly).

HELP:

Other than the instructor's office hours, there are other places to obtain assistance:

- **CS Lab tutors** (good for help on Programming Assignments):

<https://cs.txstate.edu/resources/labs/tutoring/>

- **CLC tutoring** (good for help on understanding the material):

<https://hlsamp.cose.txstate.edu/clc/clc-tutoring.html>

- **If you received a C grade in CS 1428 I expect you to attend one SI session per week.**

Withdrawals/drops:

You must follow the withdrawal and drop policy set up by the University:

<http://www.registrar.txstate.edu/registration/dropping-or-withdrawing.html>

For rest of the deadlines regarding withdrawal please use university calendar

<https://www.registrar.txst.edu/registration/ac/academic-calendar.html>

Accommodations for students with disability:

Any student with needs requiring special accommodations should contact the office of disability services at the LBJ student center (5-5.1). Students who qualify for extra time for exams must take their test with ATSD and must schedule their test at the same time the test is given in class. Note: you must submit your request with ATSD at least 2 business days before the exam date!

Statements from the University on:

Classroom Civility, measures for health, wellness, and safety, the Bobcat Pledge, and compliance with health standards: See course home page on Canvas for these.